

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM

August 2023

ALGEBRA (Ph.D. Version)

Instructions to the student

- a. Answer all six questions; each will be assigned a grade from 0 to 10.
 - b. Use a different booklet for each question. Write the problem number and your **code number (not your name)** on the outside of the booklet.
 - c. Keep scratch work on separate pages in the same booklet.
-

1. Let G be a finite group and let Z be the center of G (that is, Z = the elements that commute with everything in G). Let p be a prime and suppose H is a normal subgroup of G of order p .
 - (a) Show that if G is a p -group, then $H \subseteq Z$.
 - (b) Give an example where G is not a p -group and H is not contained in Z .
2. Let V be a finite-dimensional complex vector space and let W be a subspace. Let T be a linear transformation of V such that $T(W) \subseteq W$. Let $m(X)$ be the minimal polynomial of T on V and let $m_1(X)$ and $m_2(X)$ be the minimal polynomials of the linear transformations induced by T on W and V/W , respectively.
 - (a) Show that $m_1(X)$ divides $m(X)$ and that $m_2(X)$ divides $m(X)$.
 - (b) Show that $m(X)$ divides $m_1(X)m_2(X)$.
 - (c) Show that if $m_1(X)$ and $m_2(X)$ are relatively prime then $m(X) = m_1(X)m_2(X)$.
 - (d) Give an example where $m(X) \neq m_1(X)m_2(X)$.
3. Let R be a ring containing the complex numbers $\mathbb{C}$, and suppose R is of dimension 2 as a $\mathbb{C}$ -vector space. Furthermore, suppose $\mathbb{C}$ commutes with all elements of R . (R is not assumed to be commutative; however, part (b) implies in fact that it is.)
 - (a) Let $b \in R$ with $b \notin \mathbb{C}$. Show that there is a polynomial $f(X) \in \mathbb{C}[X]$ of degree 2 such that $f(b) = 0$.
 - (b) Show that, as rings, either $R \simeq \mathbb{C}[X]/(X^2)$ or $R \simeq \mathbb{C} \oplus \mathbb{C}$ (with componentwise operations).

CONTINUED ON NEXT PAGE

4. Let R be a commutative ring with 1 and let M be an R -module. Recall that M is *injective* if whenever there is a diagram of R -modules

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \xrightarrow{f} & B & \xrightarrow{g} & C \longrightarrow 0 \\ & & \downarrow \alpha & & & & \\ & & M & & & & \end{array}$$

with exact top row, there is an R -module homomorphism $\beta : B \rightarrow M$ such that $\beta \circ f = \alpha$.

- (a) Show that if A is injective and $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ is an exact sequence of R -modules, then $B \simeq A \oplus C$.
 - (b) Suppose R is an integral domain with field of fractions F . Show that if R is injective as an R -module, then $R = F$.
5. Let $K/\mathbb{Q}$ be a Galois extension with Galois group G .
- (a) Show that if $G \simeq A_4$, the group of even permutations of 4 objects, then there is a polynomial $f(X) \in \mathbb{Q}[X]$ of degree 4 such that K is the splitting field of $f(X)$.
 - (b) Show that if G is abelian of order n and K is the splitting field of an irreducible polynomial $g(X) \in \mathbb{Q}[X]$, then $g(X)$ must have degree n .
6. Let G be a group and let H be a *normal* subgroup. Let $a_1H, \dots, a_nH$ be the cosets of H . Let V be the vector space over $\mathbb{C}$ with basis $\{e_1, \dots, e_n\}$. For $g \in G$, define the linear transformation T_g by $T_g(e_i) = e_j$ if $ga_iH = a_jH$. The map

$$\rho : G \longrightarrow GL(V), \quad g \longmapsto T_g$$

gives a representation of G (you do not need to show this). Show that when ρ is decomposed as a direct sum of irreducible representations, the trivial representation of G appears exactly once. (*Hint:* Show that $gaH = aH \iff g \in H$.)